



Sonu Kumar

Bachelor of Technology

Electrical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., EE	National Institute of Technology, Patna	7.95	2023-2027
Senior Secondary	BSEB	91.2	2022
Secondary	CBSE	88.6	2020

INTELLECTUAL PROPERTY RIGHT (IPR)

• A Line-Marking Robot

[View](#)

An autonomous, high-precision line-marking robot

- An autonomous, high-precision line-marking robot designed to automatically draw sports field and ground markings using coordinate-based navigation and sensor fusion.

PROJECTS

• TourKaro – AI-Powered Tourist Guide Platform

Jul 2025 – Present

FastAPI, Node.js, MongoDB, Redis, Gemini API, ChromaDB, RAG, Docker, JWT, CI/CD

[Website Link](#)

- Engineered a production-grade AI travel platform leveraging **RAG (Retrieval-Augmented Generation)**, **vector embeddings**, and **FAISS** for context-aware destination recommendations and intelligent trip planning.
- Developed scalable **FastAPI** and **Node.js** microservices with secure **JWT authentication**, REST APIs, and modular architecture supporting real-time tourist assistance.
- Implemented **Redis-based caching** and optimized database queries, significantly reducing API latency and improving system responsiveness under concurrent user loads.
- Integrated **Google Gemini API** with semantic search pipelines to provide personalized travel guidance, itinerary generation, and conversational AI support.
- Containerized services using **Docker** and deployed on cloud infrastructure with automated **CI/CD pipelines**, ensuring reliable and maintainable production releases.

• House Price Prediction Web App

[GitHub Repo](#)

Flask, Python, HTML/CSS/JS, Machine Learning

- Built and deployed a Flask-based prediction service exposing **REST APIs** for real-time inference and model management.
- Engineered a complete ML pipeline including data preprocessing, model training, evaluation, serialization, and API-based inference delivery.

SKILLS

- **Programming**: Python, C++, C, JavaScript
- **Backend & Databases**: FastAPI, REST APIs, MCP, LangChain RAG, MongoDB, FAISS, Redis
- **Tools & Platforms**: Pandas, NumPy, PyTorch, Matplotlib, Git, GitHub, Docker, Linux, CI/CD
- **Hardware & IoT**: ESP32, Arduino, Raspberry Pi, Jetson Nano

CERTIFICATIONS

• IIT Patna, DroneX – The Drone Challenge

Jun 2024

• NIT Patna, Laser Bot Race Challenge

Jan 2024

ACHIEVEMENTS

- Ranked 3rd in district, Intermediate Science (2022)
- Won Gold in Laser Bot Race, Silver in Tinkering Lab, and Bronze in Grand Line Challenge

POSITIONS OF RESPONSIBILITY

• Vice President, Robotics Club, NIT Patna

2025 – Present

• Technical Team Member, Robotics Club, NIT Patna

2023 – 2025